

KAIJIE WANG

E-mail: kjwang000@hotmail.com, Phone: +86 18758285933

No. 3, Lane 245, Yaoyuan Road, Pudong New Area
Shanghai 200126, China

I. EDUCATION

University of Chinese Academy of Sciences, joint training with
Shanghai Advanced Research Institute, Chinese Academy of Sciences

Beijing, China
Shanghai, China
June 2024

Ph.D. in Communication and Information Systems (Outstanding Graduate Award)

Research interests: Wireless communications, non-orthogonal multiple access (NOMA), adaptive modulation and coding, energy-efficient resource allocation, reconfigurable intelligent surfaces (RIS)

Dissertation: “Research on Key Technologies of Wireless Resource Allocation for Uplink Non-Orthogonal Multiple Access”

Summary of Dissertation: This dissertation addresses the diverse service requirements of 6G networks, aiming to enhance throughput, energy efficiency, and user fairness in uplink NOMA systems through advanced techniques and theories. Specifically, it explores adaptive modulation and coding, fairness-aware energy-efficient power allocation, and resource allocation for uplink NOMA systems aided by simultaneously transmitting and reflecting reconfigurable intelligent surfaces. Numerical results show significant performance improvements of the proposed algorithms compared to traditional methods.

Supervisors: Prof. Honglin Hu and Prof. Ting Zhou

College of Information Science and Electronic Engineering,
Zhejiang University

Hangzhou, China
June 2019

B.E. in Electronic Science and Technology

Relevant coursework: Digital signal processing, digital image processing, basic of machine learning, deep learning, basic of data mining, data analysis and algorithm design, information theory

Thesis: “Sparse Coding of Deep Learning in Target Classification”

II. PUBLICATIONS

1. **K. Wang**, T. Zhou, T. Xu, H. Hu, and X. Tao, “Asymmetric Adaptive Modulation for Uplink NOMA Systems,” in *IEEE Transactions on Communications*, vol. 69, no. 11, pp. 7222-7235, Nov. 2021.
Abstract: The asymmetric channels in uplink NOMA systems can bring distinct uncertainty of the bit error rate and throughput performances to the users. To this end, we propose an asymmetric adaptive modulation algorithm to dynamically select modulation schemes and improve the sum throughput for uplink NOMA systems.
2. **K. Wang**, T. Zhou, T. Xu, and H. Hu, “Error-Compensated Adaptive Modulation and Coding for Uplink NOMA Systems,” in *Proc. VTC2022-Spring*, Helsinki, Finland, 2022, pp. 1-5.
Abstract: Current adaptive modulation and coding (AMC) schemes may not be preferable for NOMA systems due to the error propagation. To address this problem, we propose an error-compensated AMC algorithm to guarantee the target block error rate and improve the sum throughput for uplink NOMA systems.
3. J. Wu, T. Xu, T. Zhou, and **K. Wang**, “Feature Detection Based Spectrum Sensing in NOMA System,” in *Mobile Networks and Management (MONAMI 2022)*, vol.418, pp.201, 2022.
Abstract: In this paper, we propose a transceiver architecture that combines the NOMA and spectrum sensing techniques to address the increasing demand for spectrum resources in future communications.

Our experiments demonstrate the feasibility of this approach, showing good detection performance.

4. **K. Wang**, T. Zhou, T. Xu, Y. Huang, H. Hu, and X. Tao, “Fairness-Aware Energy-Efficient Power Allocation for Uplink NOMA Systems With Imperfect SIC,” in *IEEE Wireless Communications Letters*, vol. 12, no. 12, pp. 2018-2022, Dec. 2023.

Abstract: We first introduce a new metric called geometric energy efficiency (GEE), then a dynamic power allocation scheme is developed by maximizing the GEE. Numerical results show that the proposed power allocation scheme achieves at least a 15.1% EE gain over benchmarks while also providing better or comparable fairness performance.

5. **K. Wang**, T. Zhou, T. Xu, and H. Hu, “Energy-Efficient Resource Allocation for STAR-RIS Aided Uplink NOMA Systems With Imperfect SIC,” as a part of Ph.D. dissertation.

Abstract: The simultaneously transmitting and reflecting reconfigurable intelligent surfaces (STAR-RIS) is a promising technology for future mobile systems. To maximize the EE/GEE of a STAR-RIS aided uplink NOMA system, we propose a resource allocation algorithm that jointly optimize the receive beamforming, phase shift, amplitude coefficients, power and time allocation.

III. SKILLS

Technology: MATLAB, C, Python, PyTorch programming, Microsoft Office suite, LaTeX

Language: English (skilled in conversation and writing), Chinese (native language)

IV. AWARDS

1. **National Scholarship**, granted by Ministry of Education of the People’s Republic of China, 2021
2. **Merit Student**, granted by University of Chinese Academy of Sciences, 2020 and 2022
3. **President Award**, granted by Chinese Academy of Sciences, 2024
4. **Outstanding Graduate of Beijing**, granted by Beijing Municipal Commission of Education, 2024
5. **New-Star of Key Technologies R&D**, granted by Chinese Academy of Sciences, 2024

V. REFERENCES

Dr. Honglin Hu

Professor and director of the Research and Development Center for Intelligent Information and Communication Technology

Shanghai Advanced Research Institute, Chinese Academy of Sciences

No. 99, Haik Road, Pudong New Area, Shanghai, China

E-mail: huhl@sari.ac.cn

Dr. Ting Zhou

Professor and deputy dean of the School of Microelectronics

Shanghai University

No. 20, Chengzhong Road, Jiading District, Shanghai, China

E-mail: zhouting@shu.edu.cn

Dr. Tianheng Xu

Professor at the Research and Development Center for Intelligent Information and Communication Technology

Shanghai Advanced Research Institute, Chinese Academy of Sciences

No. 99, Haik Road, Pudong New Area, Shanghai, China

E-mail: xuth@sari.ac.cn